

Inventra – Intelligent Inventory Management System

Overview of



Module 1 – Authentication & Access Control
Module 2 – Product / Inventory Management
Module 3 – Alerts & Monitoring System



INTRODUCTION

1. **Modern applications are divided into independent modules**
2. **Each module performs a specific responsibility**
3. **Modular design improves security, maintainability, and scalability**
4. **All modules work together to form a complete intelligent system**



MODULE OVERVIEW

- 1. Module 1: Controls system access using authentication and roles**
- 2. Module 2: Manages products, inventory, and stock calculations**
- 3. Module 3: Monitors system conditions and generates alerts**



MODULE 1 – AUTHENTICATION & ACCESS CONTROL

➤ MODULE 1 – PURPOSE

- 1. Authenticate users using username and password**
- 2. Ensure only valid users can log in**
- 3. Restrict system access based on user roles**
- 4. Protect the system from unauthorized usage**



MODULE 1 – SCENARIO

➤ Scenario: Role-Based Login & Access Control

1. User provides login credentials
2. System validates username and password
3. User role (ADMIN / USER) is identified
4. Access permissions are applied based on role



MODULE 1 – AUTHENTICATION CODE

```
String username = "admin";  
String password = "admin123";  
  
if (username.equals("admin") && password.equals("admin123")) {  
|   System.out.println("Login Successful");  
} else {  
|   System.out.println("Invalid Login");  
}
```

MODULE 1 – AUTHORIZATION CODE

```
String role = "ADMIN";

if (role.equals("ADMIN")) {
|   System.out.println("Full Access Granted");
}
else if (role.equals("USER")) {
|   System.out.println("Limited Access Granted");
}
else {
|   System.out.println("Unauthorized Access");
}
```



MODULE 1 – KEY POINTS

- 1. Authentication verifies user identity**
- 2. Authorization controls user permissions**
- 3. Invalid login attempts are handled safely**
- 4. Unauthorized roles are denied access**



MODULE 2 – PRODUCT / INVENTORY MANAGEMENT

MODULE 2 – PURPOSE

- 1. Store product details in the system**
- 2. Maintain accurate inventory records**
- 3. Handle stock received in multiple shipments**
- 4. Display current available stock at any time**



MODULE 2 – SCENARIO

► Scenario: Sports Equipment Store Inventory System

1. Store sells multiple sports items
2. Each item has a unique product code
3. Stock arrives from suppliers at different times
4. Total stock is calculated from all shipments



MODULE 2 – PRODUCT ENTITY & SHIPMENT HANDLING

```
class Product {  
    String productCode;  
    String productName;  
    int totalStock = 0;  
}  
  
void addShipment(int quantity) {  
    totalStock += quantity;  
}
```



MODULE 2 – INVENTORY UPDATE & STOCK DISPLAY

```
Product bat = new Product();  
bat.addShipment(20);  
bat.addShipment(15);  
  
System.out.println("Product Code: CRB101");  
System.out.println("Available Stock: " + totalStock);
```



MODULE 2 – KEY POINTS

- 1. Unique product code ensures correct identification**
- 2. Multiple shipments are handled efficiently**
- 3. Total stock is calculated dynamically**
- 4. Manager can check inventory status anytime**



MODULE 3 – ALERTS & MONITORING

MODULE 3 – PURPOSE

- 1. Monitor system conditions continuously**
- 2. Detect warning and critical situations**
- 3. Generate alerts to notify users**
- 4. Support timely decision-making**



MODULE 3 – SCENARIO

Scenario: Mobile Data Usage Alert System

- 1. Users have limited data plans**
- 2. System tracks data usage continuously**
- 3. Alerts are generated when data is low or exhausted**



MODULE 3 – ALERT LOGIC CODE

```
int dataLimit = 100;
int dataUsed = 92;
int remaining = dataLimit - dataUsed;

if (remaining <= 10 && remaining > 0) {
    System.out.println("Alert: Data Running Low");
}
```


MODULE 3 – NORMAL & EXCEEDED CODE

```
else if (remaining <= 0) {  
|   System.out.println("Alert: Data Limit Exceeded");  
}  
else {  
|   System.out.println("Data Usage Normal");  
}
```



MODULE 3 – ALERT PRIORITY

- 1. Alerts are handled using Stack (LIFO)**
- 2. Most recent alert is shown first**
- 3. Resolved alerts are removed from the system**
- 4. Reduces alert confusion and noise**



MODULE INTEGRATION

- 1. Module 1 secures system access**
- 2. Module 2 manages core business data**
- 3. Module 3 monitors system health**
- 4. Together they form a complete intelligent system**



CONCLUSION

1. **Role-based security ensures safe access**
2. **Inventory module maintains accurate stock**
3. **Alert module provides timely notifications**
4. **System design matches real-world applications**



THANK YOU